SPY Volatility Managed

Strategy report · Research desk · 26 October 2026

Status: passed, confirms test zone · *full-sample t 3.33*

Abstract

SPY Volatility Managed holds SPY each month at a weight that falls when the previous month's realised variance was high and rises when it was low, capped at 1.5 times capital, with the rest in T-bills. Over February 2005 to September 2026 it earns a net Sharpe ratio of 0.66 (90 % bootstrap range 0.33 to 0.99). On the test, January 2021 to September 2026, the net Sharpe is 0.82 (range 0.20 to 1.45), against 0.78 for SPY over the full sample. **Three findings weigh against it: the book is SPY with a beta of 0.90 over the full sample and 1.15 on the test, and its alpha over SPY alone is 3.1 % a year (t 0.96) on the full sample and −1.2 % (t −0.31) on the test; the best 1 % of days make 102 % of the net P&L; and the book lost 52.9 % from July 2007 to March 2009.** The sheet is passed, on paper only.

1 Hypothesis and method

Investors who stay fully exposed through months of high variance are not paid for the extra risk, because the return per unit of variance falls when variance is high; scaling exposure by the inverse of last month's variance harvests that. The source is Moreira and Muir (2016), measured in-sample on US equity factors over decades, before costs and without a leverage cap. The authors call the explanation speculative, so the payer is the desk's hypothesis. A rival reading is that the rule is a lower-beta SPY, a single avoided crash in 2008, or a trend bet, which is why the sheet compares the book with SPY and two trend twins.

The universe is a single instrument, SPY. Each month the weight is the constant c divided by the realised variance of the previous month's daily returns, capped at 1.5, with the rest in T-bills at the TB3MS rate; c was fixed on the train so that the managed volatility equals SPY's. Trades take place at the first close of the next month. The 0.5 above 1 is borrowed at the T-bill rate plus 50 bp, a rate with no source, and trading costs are 2 bp. The train runs from 2005 to 2017, the validation from 2018 to 2020 and the test from January 2021; the test was opened once, on 26 October 2026.

2 Performance

The full sample includes the train, on which the strategy was designed; the test is the honest estimate. Over the test the strategy returned 16.5 % a year for 20.2 % volatility, a net Sharpe of 0.82 whose bootstrap range runs from 0.20 to 1.45, wide enough to contain the full-sample 0.66. By period the net Sharpe is 0.64 in the train, 0.44 in the validation and 0.82 in the test. The net return was −33 % in 2008, 43 % in 2013, −24 % in 2022 and 41 % in 2021. The maximum drawdown, −52.9 %, runs from July 2007 to March 2009, was recovered in December 2010 and kept the book under water for 1239 days.

	Sample	5th percentile	Median	95th percentile
Annualised return	13.1 %	6.7 %	13.1 %	19.1 %
Annualised volatility	19.9 %	18.8 %	19.9 %	21.1 %
Sharpe ratio	0.66	0.33	0.66	0.99
Maximum drawdown	−52.9 %	−81.9 %	−47.0 %	−31.3 %
Worst month	−12.7 %	−30.8 %	−19.0 %	−13.3 %
Positive months	63 %	58 %	63 %	68 %

Table 1: Net performance, full sample, February 2005 to September 2026, after costs and financing. Percentiles of a stationary block bootstrap of daily returns (blocks of 20 days, 2 000 paths). Returns in fraction of capital, not compounded.

	Sample	5th percentile	Median	95th percentile
Annualised return	16.5 %	4.1 %	16.4 %	28.3 %
Annualised volatility	20.2 %	18.7 %	20.1 %	21.8 %
Sharpe ratio	0.82	0.20	0.81	1.45
Maximum drawdown	−30.4 %	−50.2 %	−28.4 %	−18.0 %
Worst month	−10.4 %	−16.9 %	−12.2 %	−8.5 %
Positive months	62 %	54 %	64 %	72 %

Table 2: Net performance over the test alone, January 2021 to September 2026. Same method as Table 1.

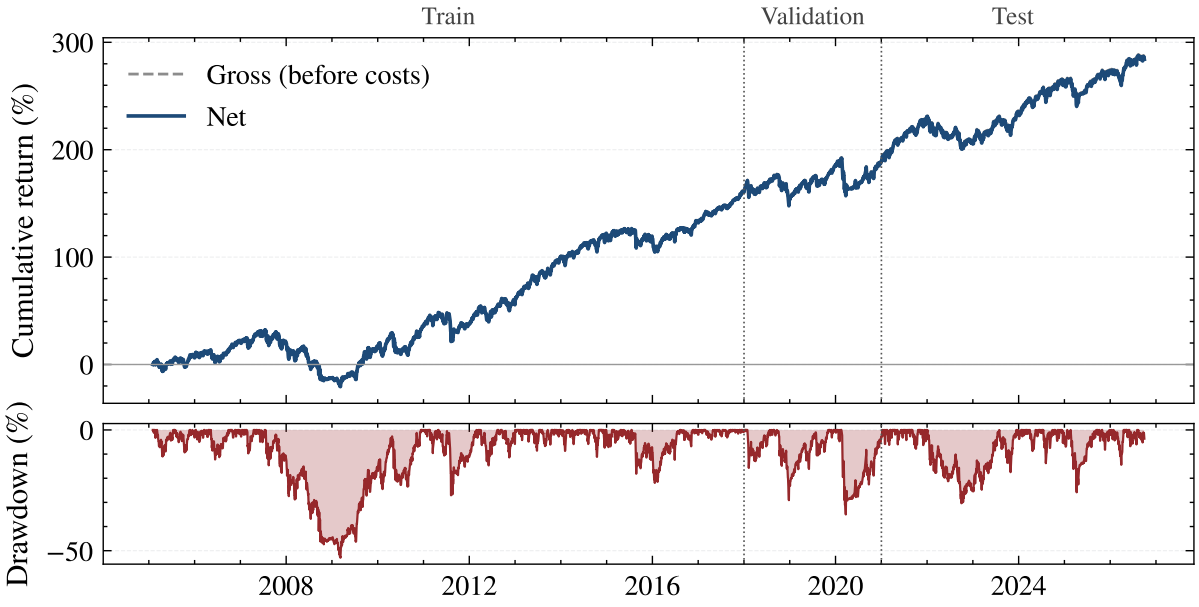


Figure 1: Cumulative net and gross return (top) and drawdown of the net return (bottom), in percent of capital, linear scale. Dotted lines mark the start of the validation and of the test.

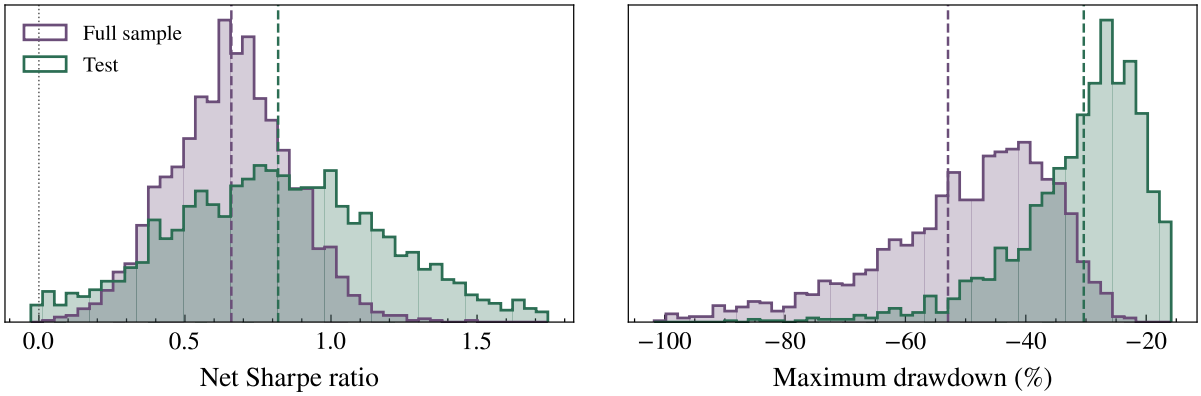


Figure 2: Bootstrap distributions of the net Sharpe ratio (left) and of the maximum drawdown (right). Dashed lines: the values measured on the sample.

3 Market exposure

The strategy is a bet on the equity market: its beta to SPY is 0.90 over the full sample (t 13.18) with a correlation of 0.86, and on the test the beta is 1.15 (t 22.89) with a correlation of 0.94. The alpha over SPY alone is 3.1 % a year on the full sample (t 0.96) and −1.2 % on the test (t −0.31), so it is not shown on either. Test.md measures the alpha over SPY and both trend twins, the comparison the sheet calls its key one: 0.196 % a month pooled over all three periods, a t of 1.95, under the bar of 2. SPY itself scores 0.78 over the full sample.

	Full sample	Test
Beta to SPY (daily)	0.90 (t 13.18)	1.15 (t 22.89)
Correlation to SPY	0.86	0.94
Alpha over SPY, annualised	3.1 % (t 0.96)	−1.2 % (t −0.31)

Table 3: Exposure to the market. Regression of daily net returns on SPY daily returns; t-statistics with Newey-West standard errors (10 lags).

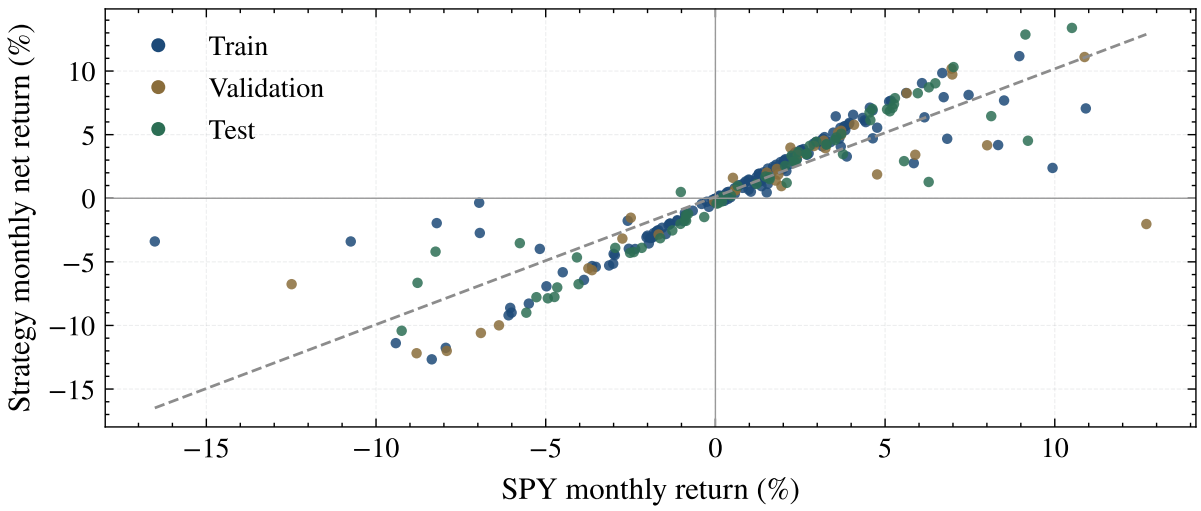


Figure 3: Monthly net return of the strategy against the monthly return of SPY. Dashed line: least-squares fit.

4 Where the profit comes from

There is no short leg: the book is one long price leg. In the train the price made 185 % of capital, financing at the T-bill rate cost 24 % and costs 1 %; in the validation the price made 36 %, financing 7 %; in the test the price made 122 %, financing 27 % and costs under 1 %. The financing line is large because the book is long and levered above 1 most of the time. The hypothesis says the timing earns the money; the P&L shows a levered long position in SPY, and the timing is not separable from it in these tables. The payer named in the hypothesis cannot be said to have paid.

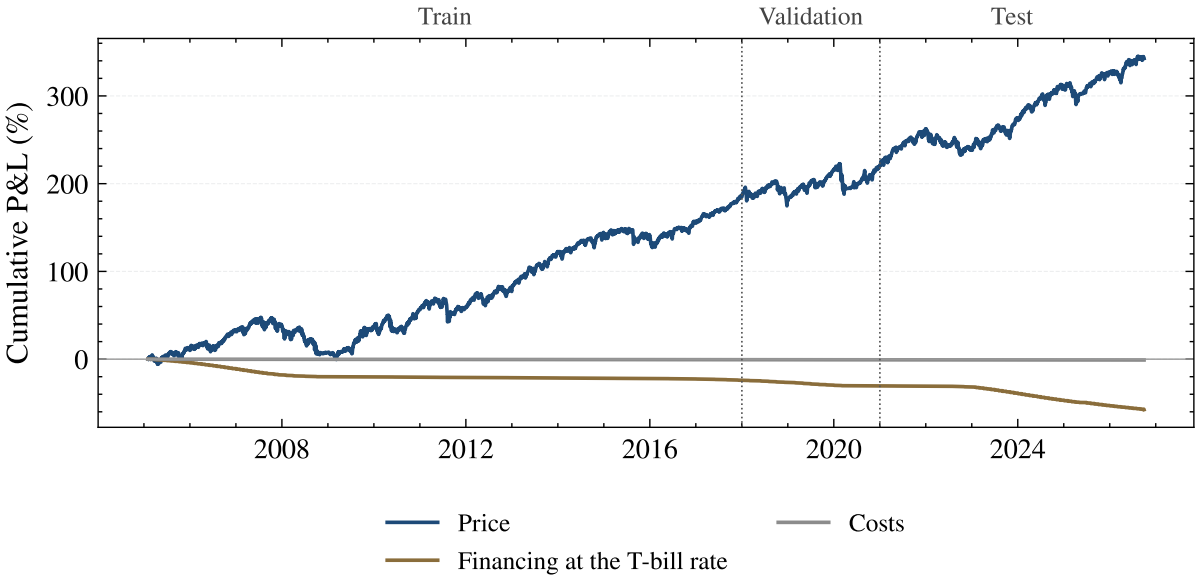


Figure 4: Cumulative P&L by source, in percent of capital: price, financing at the t-bill rate, and costs.

5 Concentration

The best 1 % of days, 80 days in the full sample, make 102 % of the net P&L: without them the Sharpe is −0.02. In the test the best 21 days make 75 % and the Sharpe without them is 0.22. On SPY itself the same cut takes the Sharpe from 0.78 to −0.01, so the thick tail is the nature of the market. Without their two best days, only 33 % of months are positive. The big days are regular: each year the best two days of each month are positive and the other days negative, and the other days lost most in 2008 (−91 %) and 2022 (−71 %).

	Full sample	Test
Net Sharpe ratio, all days	0.66	0.82
Share of net P&L from the best 1 % of days	102 % (80 days)	75 % (21 days)
Net Sharpe ratio without the best 1 % of days	−0.02	0.22
Annualised return without the best 1 % of days	−0.3 %	4.1 %
Net Sharpe ratio without the best 5 % of days	−1.87	−1.68
SPY Sharpe ratio, all days / without its best 1 %	0.78 / −0.01	

Table 4: Concentration of the gains in time. A day removed is set to zero, as if the strategy had been flat that day. The SPY line gives the same cut on the market over the same span.

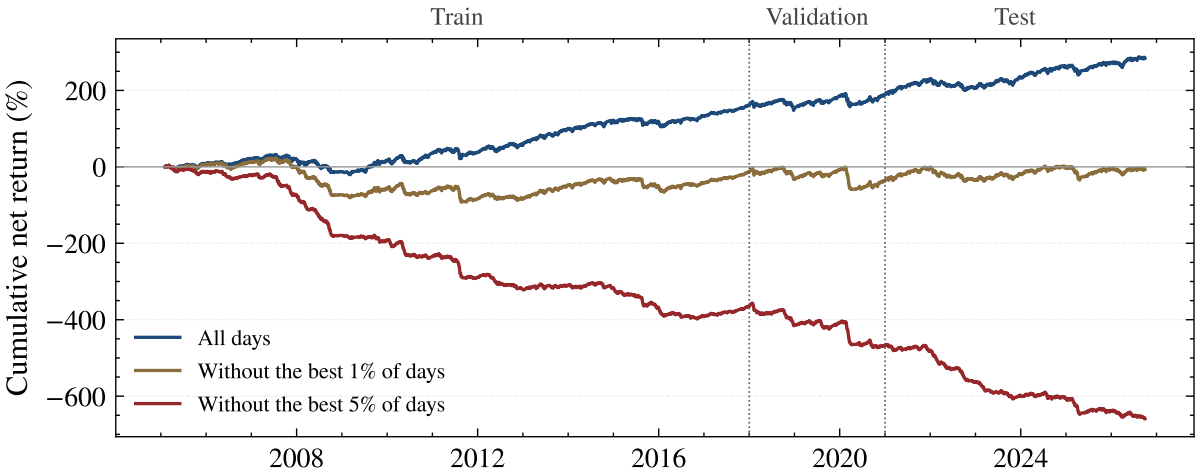


Figure 5: Cumulative net return with all days, and with the best 1 % and 5 % of days set to zero.

6 Implementation

The book holds one name, long, at an average gross exposure of 1.33 in the train, 1.23 in the validation and 1.36 in the test. Turnover is 1.98 a year in the train, 3.77 in the validation and 1.97 in the test, and the cost drag is under 0.1 % a year. The break-even cost is 628 bp in the train, 257 bp in the validation and 839 bp in the test against the 2 bp assumed, so costs are not the issue. The borrow rate above 1x is the weak assumption: it is unsourced, and financing took 27 % of capital in the test.

	Train	Validation	Test
Names held long / short, average	2 / 0	2 / 0	2 / 0
Gross exposure, average	1.33	1.23	1.36
Holding period of a position, days	669	236	789
Turnover, times the capital a year	2.0	3.8	2.0
Cost drag, % of capital a year	−0.0 %	−0.1 %	−0.0 %
Break-even cost, bp one way	628.0	256.9	838.7

Table 5: Implementation. The break-even cost is the one-way cost per unit traded at which the gross P&L of the period would be fully spent.

7 Robustness

The frozen version scores a net Sharpe of 0.60 over train and validation. A fix of the 2008-10-13 return gives 0.60, costs doubled 0.60, a spread of 150 bp 0.58, trades one day late 0.57 and two days late 0.57. The 3-month variance variant scores 0.47. The Sharpe level is robust to every stress run. The level is not the question: the alpha over SPY and the trend twins fell with a lag of one or two days, which the net Sharpe does not show.

	Train	Train and validation	Δ train	Δ train and validation
Frozen version, as tested	0.65	0.60		
base-fix1013	0.65	0.60	+0.00	−0.00
Borrow spread 100 bp	0.64		−0.01	
Borrow spread 150 bp		0.58		−0.02
Cost 4 bp		0.60		−0.00
Borrow spread 25 bp		0.60		+0.01
Trades 2 days later		0.57		−0.03
Trades 1 day later		0.57		−0.03

Table 6: Stress tests of the frozen version, run by the Red team before the test: net Sharpe ratio, and its change from the frozen version on the same period.

	Kind	Run on	Net Sharpe	Frozen version
variant1	variant	Train and validation	0.47	0.60

Table 7: Other versions run before the test, each on the period where the desk ran it: variants considered at the validation, controls and diagnostics that check the mechanism. They are not the strategy; the last column gives the frozen version on the same period for comparison.

8 Luck and multiple testing

Two tries are counted for this strategy; its family and asset (low-vol, equities) have spent 2 tries with an effective number of 1.0, and the luck line is −0.21, against 0.76 for the whole desk after 38 tries. The train net Sharpe of 0.64 minus the line gives 0.85, and the probability that the true train Sharpe exceeds the line is 100 %. A negative line means the history is long and the trial count is low, so it only says the level is not luck; it says nothing about the alpha over SPY, which is the open question.

	Value
Tries counted for this strategy	2
Tries in the family, low-vol on equities (effective number)	2 (1.0)
Luck line: best Sharpe of 1.0 tries with no skill, 14.4 years	-0.21
Train net Sharpe ratio / minus the luck line	0.64 / 0.85
Probability that the true train Sharpe exceeds the luck line	100 %
Whole desk: tries (effective number) / luck line	38 (9.6) / 0.76

Table 8: Multiple testing. Luck line: expected best annual Sharpe ratio of N_{eff} independent tries with no skill (Bailey and López de Prado, 2014); N_{eff} discounts tries by their correlation before costs. The probability is the deflated Sharpe ratio, corrected for the skewness (−0.7) and the kurtosis (11) of daily train returns.

9 Risks and limits

The book is levered SPY, so it loses what SPY loses: −52.9 % in 2007 to 2009 and −24 % in 2022. The 2020-03 month had a weight of 0.77 while SPY fell, a reminder that last month's variance does not anticipate a crash. The alpha depends on trading at the first close of the month: with a delay it fades. The borrow rate above 1x is unsourced, and the 2008-10-13 SPY return in the data has not been checked on a second source. The test 2021-2026 is a strong equity market, which flatters a levered long book.

10 Decision

The sheet is passed, on paper only, per Test.md of 26 October 2026: the zone confirms (test alpha 0.366 % a month against 0.175 % earlier) and no kill fires under the Kill Policy. The money rule asks a full-sample t of 2 or more and a test that does not contradict; the level meets it (full-sample net t 3.33), but the final proof of Pass 1, the alpha over SPY and both twins at t 2, stays at 1.95, so the sheet is paper only. The decision would change if the alpha over SPY and the twins reaches t 2 on further data outside the sample, or if the borrow rate is sourced and the result holds at it.

A Monthly returns

	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec	Year
2005		2.1	−3.0	−3.1	4.5	−0.1	5.4	−1.8	0.8	−4.0	6.1	−0.7	**6.2**
2006	3.1	0.4	2.0	1.3	−5.1	−0.2	0.1	2.7	3.4	4.1	2.3	1.4	**15.3**
2007	1.6	−3.6	1.1	6.0	4.4	−2.8	−5.3	1.3	3.3	1.0	−6.4	−1.8	**−1.2**
2008	−8.6	−1.8	−1.4	5.5	0.5	−12.7	−1.2	1.1	−11.4	−3.4	−0.4	0.7	**−32.9**
2009	−2.0	−3.4	4.2	2.4	2.8	−0.2	8.1	4.1	6.4	−3.1	6.4	2.3	**27.8**
2010	−5.3	4.5	9.0	2.3	−11.8	−4.0	4.7	−5.8	11.2	5.7	−0.0	9.8	**20.3**
2011	3.5	5.2	−0.0	4.3	−1.7	−2.5	−2.9	−8.3	−2.7	7.1	−0.4	0.5	**1.9**
2012	4.7	6.3	4.8	−1.0	−9.0	6.6	1.4	3.8	3.8	−2.7	0.8	1.3	**20.7**
2013	7.6	1.9	5.6	2.8	3.5	−2.0	7.7	−4.4	4.7	6.9	4.4	3.9	**42.7**
2014	−5.4	7.1	1.2	1.0	3.5	3.1	−2.0	5.9	−2.1	3.6	3.5	−0.3	**19.1**
2015	−4.5	8.3	−2.3	1.5	1.9	−3.1	3.3	−9.2	−4.0	7.7	−0.1	−2.6	**−3.0**
2016	−6.9	−0.1	7.9	0.5	2.5	0.5	4.7	0.1	−0.0	−2.7	5.5	3.0	**15.0**
2017	2.6	5.8	0.2	1.4	2.0	0.9	3.0	0.3	2.9	3.4	4.4	1.7	**28.6**
2018	8.3	−5.7	−3.2	1.6	3.4	0.7	5.3	4.5	0.6	−10.6	1.9	−12.2	**−5.3**
2019	4.2	3.9	2.3	5.8	−10.0	10.2	2.0	−2.8	1.0	4.0	5.2	4.1	**29.7**
2020	−0.2	−12.0	−6.8	−2.0	1.9	1.4	3.4	9.7	−5.5	−1.5	11.1	5.0	**4.5**
2021	0.5	4.1	6.7	7.9	1.0	3.3	3.6	4.4	−7.0	10.3	−1.2	7.0	**40.7**
2022	−7.8	−3.9	3.5	−6.7	0.1	−4.2	4.5	−4.7	−10.4	6.5	2.9	−3.5	**−23.6**
2023	8.7	−4.3	5.1	1.5	0.2	9.0	4.2	−3.1	−7.8	−3.9	12.9	6.1	**28.7**
2024	1.7	7.1	4.2	−6.8	7.0	4.6	1.1	2.9	1.2	−1.7	8.3	−4.2	**25.3**
2025	3.5	−2.5	−9.0	−1.3	1.3	6.8	2.9	2.5	4.8	3.0	−0.2	−0.4	**11.3**
2026	1.7	−1.8	−7.9	13.4	7.4	−2.0	−0.4	3.5	−1.5				**12.4**

Table 9: Monthly net return, % of capital.

Runs: 20261025-142825-train-base, 20261025-160847-train+validation-base, 20261026-085632-test-base. Strategy file: spy_volatility_managed.py. Frozen version: base.